

Sprint 6 — Frontend Testing Guide

Tests for the Quotation Generation and Approval/Conversion UI — companion to the Sprint 6 Frontend Build Guide.

repo: frontend

test runner: node --test

style: source-inspection (no jsdom/RTL)

ON THIS PAGE

1. The testing convention this repo actually uses
2. GitHub issue
3. Branch
4. Claude Code master prompt
5. Commit messages
6. Pull request
7. Definition-of-Done checklist

01 The testing convention this repo actually uses

IMPORTANT — DON'T ASK CLAUDE CODE FOR JEST OR REACT TESTING LIBRARY

This is the one place this frontend guide most needs to diverge from the backend Testing Guide PDF. The backend uses Jest with real database integration tests. This frontend has no Jest, no jsdom, and no `@testing-library/react` installed — `package.json`'s `test` script is just `node --test`. Every existing test (`test/finance-dashboard.test.mjs`, `test/analytics-dashboard.test.mjs`, `test/dashboard-layout.test.mjs`) reads a component's raw source with `readFileSync` and asserts with regex — confirming specific strings, endpoint paths, and UI copy exist, not that the component actually renders correctly in a browser. Treat these as "contract" tests, not real rendering tests, and follow that exact pattern below. If a prompt asks for Jest/RTL, it will fail to install and break CI.

This also means these tests cannot catch runtime bugs (a broken `useEffect`, a bad prop type at render time). That's a real limitation worth calling out, not papering over — the manual verification steps in both PRs and the demo script in the Build Guide are what actually prove the feature works, on top of these tests confirming the code contains what it's supposed to.

02 GitHub issue

● ISSUE BODY

Title: Tests – Quotation Generation & Approval UI (UC-03 / UC-04)

Labels: epic:quotation, epic:frontend, epic:testing

Sprint: Sprint 6

Estimate: 4 hours

User Story

As the frontend developer for the quotation feature, I want source-level tests for the quotation UI, so that regressions in the create/approve flow are caught by CI the same way Finance and Analytics already are.

Background / Context

- Builds on Issues #1 and #2 (Quotation Generation UI, Quotation Approval & Project Conversion UI) – both must be merged first.
- Test convention: node's built-in test runner (node --test), source read via `readFileSync` and asserted with regex, following `test/finance-dashboard.test.mjs` exactly. No `jsdom`/RTL is installed – do not add it for this issue.

Acceptance Criteria

AC1: Placeholder is gone

Given `src/app/dashboard/quotations/page.tsx`

When its source is inspected

Then it references `QuotationsDashboardClient` and does NOT reference `DashboardPlaceholderPage`

AC2: Required UI states are present

Given `QuotationsDashboardClient.tsx` and `quotationUtils.ts`

When inspected

Then source contains the empty-state copy, the unavailable-state copy, and the validation error copy (empty items, missing lead)

```
### AC3: Correct endpoints are wired
Given QuotationsDashboardClient.tsx
When inspected
Then it calls apiClient.post("/quotations", ...),
    apiClient.get(`/quotations/${id}`), and
    apiClient.patch(`/quotations/${id}/approve`)

### AC4: Approval edge cases have visible copy
Given QuotationsDashboardClient.tsx and quotationUtils.ts
When inspected
Then source contains distinguishable copy/branches for: the CONVERTED
    success state (projectId shown), the REJECTED disabled state, the
    already-converted informational state, and a retryable error state

### AC5: CI stays green
Given this PR
When npm run test runs in CI
Then all new and existing tests pass with 0 failures

## Technical Notes
- New file: test/quotation-dashboard.test.mjs
- Mirror test/finance-dashboard.test.mjs's structure exactly: read()
  helper, one test() block per concern, regex assertions against raw
  source strings

## Definition of Done
- [ ] npm run test passes with 0 failures (new + existing tests)
- [ ] npm run lint and npm run format:check pass
- [ ] PR raised against develop; at least 1 peer review; CI green
- [ ] CHANGELOG.md updated under [Unreleased]
- [ ] Issue linked in commit (Closes #<this-issue>)

## Estimate
4 hours

## Dependencies
Blocked by: Issues #1 and #2 (Quotation Generation UI, Quotation
Approval & Project Conversion UI) merged to develop.
```

03 Branch

● TERMINAL

```
git checkout develop
git pull origin develop
```

04 Claude Code master prompt

● PROMPT — WRITE THE TESTS

You are writing source-inspection tests for the quotation UI in the ConstructPro ERP frontend. This repo's test script is `node --test` - there is no Jest, no jsdom, no React Testing Library, and you must not add any of them. Every existing test in `test/*.test.mjs` (read `test/finance-dashboard.test.mjs` first) works by reading a component's raw source with `readFileSync` and asserting with regex via `node:assert/strict`. Follow that exact pattern.

FIRST - read these files before writing anything:

- `test/finance-dashboard.test.mjs` (the pattern to copy)
- `src/app/dashboard/quotations/page.tsx`
- `src/components/dashboard/quotations/QuotationsDashboardClient.tsx`
- `src/components/dashboard/quotations/quotationUtils.ts`
- `src/types/quotation.ts`

TASK - create `test/quotation-dashboard.test.mjs` with these `test()` blocks:

1. "quotations page renders the dedicated quotation dashboard client"
 - `assert.match` on `page.tsx` source for `/QuotationsDashboardClient/`
 - `assert.doesNotMatch` for `/DashboardPlaceholderPage/`
2. "quotation dashboard client covers the required UI states"
 - `assert.match` on `QuotationsDashboardClient.tsx` / `quotationUtils.ts` for the actual `empty-state`, `unavailable-state`, and `validation-error` copy you find in those files (use the real strings, don't guess)
3. "quotation dashboard client calls the correct quotation-service endpoints"
 - `assert.match` for `apiClient.post("/quotations", ...`
 - `assert.match` for a template literal hitting ``/quotations/${...}``
 - `assert.match` for a template literal hitting ``/quotations/${...}/approve`` via `apiClient.patch`
4. "quotation dashboard client handles the approval edge cases from BR-10.2"
 - `assert.match` for `copy/branches` covering: `CONVERTED + projectId` shown, `REJECTED` disabled state, an `already-converted` informational (non-error) path, and a `retryable error` state for downstream failures - use the real strings from the component, do not invent copy that isn't actually there

5. "quotation utils expose the expected helpers"

- `assert.match` on `quotationUtils.ts` for the real exported function names you found (`validateQuotationForm`, `normalizeQuotations`, `isQuotationUnavailableError`, `canCreateQuotation`, `canApproveQuotation`, etc.) - list only the ones that actually exist

IMPORTANT RULES:

- Read the actual files first - assert against real strings, not assumed ones. If a piece of copy or a helper name doesn't match what Issue #1/#2 actually produced, use what's really there.
- Do not install or import `jest`, `@testing-library/react`, or `jsdom`.
- Produce a complete, runnable file with all imports at the top - no placeholders or TODOs.
- Run `npm run test` after writing the file and fix anything red before showing me the result.

05 Commit messages

• COMMIT

```
git add test/quotation-dashboard.test.mjs
```

```
git commit -m "test(quotation): add source-level tests for quotation UI (UC-03/UC-04)"
```

Covers: placeholder removal, required UI states (empty/unavailable/validation), correct quotation-service endpoints (create, get-by-id, approve), BR-10.2 approval edge cases (rejected, already-converted, retryable failure), and `quotationUtils` helper surface.

Follows the existing node `--test + readFileSync + regex` convention from `test/finance-dashboard.test.mjs` - no new test framework added.

Closes #NN"

06 Pull request

• TERMINAL

```
git push origin feature/NN-quotation-ui-tests
# GitHub → Pull Requests → New pull request → base: develop
```

● PR DESCRIPTION

What this PR does

Adds source-level tests for the Quotation Generation and Approval UI (Issues #1 and #2), following the same node `--test + readFileSync + regex` convention already used for Finance and Analytics - no new test framework introduced.

Tests added (test/quotation-dashboard.test.mjs)

- Placeholder page import is gone from the quotations route
- Empty / unavailable / validation-error UI states are present
- Correct quotation-service endpoints are called (create, get-by-id, approve)
- BR-10.2 edge cases have distinguishable copy: rejected, already-converted (informational, not error), retryable failure
- quotationUtils exposes the expected helper functions

Known limitation (flagged, not hidden)

These are source-inspection tests, not rendered-component tests - there is no jsdom/RTL in this repo. They catch regressions in wiring, copy, and endpoint calls, but not runtime rendering bugs. Manual verification steps in the Build Guide's PRs are still required before merging either feature PR.

How to run

`npm run test`

Checklist

- [x] `npm run test` passes with 0 failures
- [x] `npm run lint / format:check` pass
- [x] No new test dependencies added
- [x] CHANGELOG.md updated under [Unreleased]

Closes #NN

07 Definition-of-Done checklist

- ☐ GitHub issue created, #NN replaced everywhere
- ☐ feature/NN-quotation-ui-tests branched from develop after Issues #1 and #2 are merged
- ☐ test/quotation-dashboard.test.mjs asserts against real source strings, not assumed ones
- ☐ No Jest, jsdom, or React Testing Library added to package.json

- ☐ `npm run test`, `lint`, and `format:check` all green locally and in CI
- ☐ PR opened against `develop` with `Closes #NN`, at least 1 peer review
- ☐ `CHANGELOG.md` updated under `[Unreleased]`

ConstructPro ERP · Sprint 6 · frontend repo · Person 2 testing guide — companion to the Sprint 6 Frontend Build Guide.